

QUESTION 1

(a) Suggest and explain which hunting method is more efficient. [3]

- **NE** method / **TT** method

For NE

- Ref to increased quantity of prey caught / eaten / AW;
- Ref to conserving energy by not using additional energy for electrical discharge;

For TT

- Ref combination of both electrical and chemical methods to increase effectiveness;
- Ref to electrical discharge causing immediate immobilization of prey;

(b) Suggest how a cocktail of proteins instead of a single protein is advantageous for the cone snail. [3]

- Ref. to conotoxins having a complementary shape to membrane bound receptor;
- Ref to higher chance of binding to a receptor/ AW;
- Ref. to, conotoxins targeting different prey / AW;

(c) Describe how insulin is able to trigger a response inside the muscle cell. [4]

- Binding of insulin to RTK ;
- Ref to autophosphorylation / AW;
- Cytoplasmic relay proteins bind and get activated triggering signal transduction pathways ;
- Ref to cellular response ;

(d)(i) Describe the effect of unmodified p-TIA on the escape response of zebrafish. [1]

Accept appropriate data quoted ;

(ii) Explain which deletion in the modified p-TIA has the greatest impact on its function. [4]

- Group 3 ;
- Ref. to proper data quoted;
- Ref. to deletion of amino acid sequence resulting in non-functional p-TIA /AW ;
- Ref. to correct consequence, e.g. unable to prevent flow of calcium into muscle cells;

(iii) Conotoxins are used to develop potential drugs for humans.
These drugs can be tested on zebrafish that are good model organisms since they share 70 percent of their genes with humans.

State why zebrafish is used. [2]

- Ref. to good representation since similar to humans ;
- Ref. to no need for human volunteer ;
- AVP (e.g lower cost etc) ;

(e) With reference to Fig. 1.3

(i) Suggest reasons for the high number of species found at the coral reefs. [2]

- Ref. to high habitat diversity / complex habitat ;
- Ref. to specialisation by different closely related species ;
- Example of specialisation ; prey type, size, feeding methods
- Ref. to lack of / reduced interspecific competition / competitive exclusion

(ii) describe and explain the predicted effect of global warming on the distribution of cone snails within the Philippines. [4]

- Quote poleward movement of cone snail species ;
- Quote appropriate temperature range ;
- Cone snail species with black shells are disadvantaged as they absorb more heat / ORA ;
- Ref to effect of increasing temp on cone snail ;
- Cone snail species can adapt to new environment or risk extinction ;

(iii) Identify one challenge of using morphology to classify cone snails. [1]

Ref. subjectivity in determining differences;

(f) (i) Suggest why it is useful to include gaps as indicated by dashes in Fig. 1.4 when aligning the nucleotide sequences. [1]

- Ref to adjusting for indels in some sequences ;

(ii) Explain what conclusions can be drawn from the data in Fig 1.4. [3]

- examples with data quoted on no. of differences in bases ;
- the smaller the number of differences the, more closely related / more recently divergence occurred / ORA;
- all the species share a recent common ancestor ;

(iii) Explain why there is insufficient evidence from Fig 1.4 to draw conclusions about the evolutionary relationships between cone snails. [2]

- the nucleotide sequences are only a small fraction of the whole genome ;
- use of figures to support ;

[Total: 30]



QUESTION 2

Fig. 2.1 shows the terminal differentiation of different cell types derived from a pituitary stem cell.

(a) State and explain the potency of the pituitary stem cell in Fig. 2.1.
[2]

- multipotent ;
- as it can only differentiate into a limited number of cell types ;

(b) Explain how one feature of stem cells enables them to be a possible preferred treatment over hGH injections. [2]

- stem cells can undergo long-term, self-renewal / proliferation by mitotic divisions;
- stem cells can differentiate into specialized cells ;

(c)(i) Suggest why mRNA was collected for this study instead of DNA. [1]

- ref. to presence of mRNA indicates that the gene(s) were expressed / transcribed / AW ;
- ref. to presence of DNA will not be able to reflect if the gene(s) were expressed / transcribed / AW ;

(ii) Describe how changing the sleep-wake pattern from pattern 1 to pattern 2 affects the number of genes expressed. [2]

- ref. to fewer genes have increased expression, during the day, during the night, and all the time for sleep-wake pattern 2 ; (*NB: all or none*)
- quoting of correct data from Table 2.1 ;

(iii) Explain how light can result in increased or decreased gene expression at certain times of the day. [3]

- ref. to any correct pair of quoted data / trend from Table 2.1 + correct conclusion on effect of light on gene expression ;
- presence of light detected by (photo)receptors, triggering signal transduction pathway ;
- resulting in cellular response of increased production of activator / repressor proteins, which bind to enhancers / silencers / resulting in a more / less stable TIC;

[Total: 10]



QUESTION 3

(a) Explain why symptoms of HIV and TB only appear at the later stages of infection. [3]

- ref. to HIV forming provirus in T_H / macrophage + *M. tuberculosis* forming granuloma / tubercles in alveolar macrophages ;
- ref. to HIV and *M. tuberculosis* remaining latent ;
- ref. to symptoms appearing only when sufficient / AW, T_H / macrophage / alveolar macrophages are damaged ;

(b) Discuss whether the data in Fig. 3.1 support this prediction. [4]

Does not support

- Much larger proportion do not have HIV / AIDS / AW ;
- Data to support ;
- Other factors may cause more deaths from TB ;

Support

- Large proportion of deaths of people with HIV / AIDS / are caused by TB ;
- Data to support ;
- Fewer people will be immunocompromised ;

(c) In healthy people, the number of T-helper cells ranges from 500 to 1200 cells per cm^3 of blood. In untreated people infected with HIV, the number of T-helper cells can decrease to below 200 cells per cm^3 of blood.

Explain how a low number of T-helper cells makes it more likely that untreated people infected with HIV will die if they are also infected with TB. [3]

- Ref. to opportunistic infections causing death ;
- Low levels of / less, cytokine, secreted / AW ;
- Two from:
 - activity of macrophages not stimulated
 - B-lymphocyte / plasma cells / humoral response, not stimulated
 - (so) lower concentrations of / less / no, antibody, produced / secreted ;
 - stimulates, T-cytotoxic / T-killer, cells ;
- fewer, bacteria killed / viruses eliminated / pathogens killed ;
- Fewer memory cells (to fight future infection) ;

[Total: 10]

QUESTION 4

(a) Compare the behavior of chromosomes in the two different types of cell division. [15]

Similarities

- In prophase for both, chromosomes condense
- In metaphase for both, chromosomes line up along the metaphase plate
- In anaphase for both, centromeres divide / forming daughter chromosomes which are pulled to opposite poles
- In telophase for both, chromosomes decondense

Differences

- homologous chromosomes pair up / crossing over in meiosis which does not occur in mitosis
- bivalents line up in meiosis while in mitosis, chromosomes line up singly at metaphase plate
- independent assortment in meiosis but not mitosis
- differences in chromosome behavior during anaphase i.e. sister chromatids do not separate in meiosis I

(b) Discuss how the meiotic cell cycle is crucial for evolution to occur. [10]

- crossing over allows for formation of new combinations of alleles
- independent assortment allows for random distribution of chromosomes
- leads to genetic variation in gametes, genotype and phenotype of offspring
- leads to genetic variation in a population
- genetic variation is necessary for natural selection
- individuals with favourable alleles best adapted to the environment are more likely to survive to sexual maturity
- leading to changes in allelic frequency over time, hence, microevolution occurs

QUESTION 5

(a) Compare the processes in which energy is released in aerobic and anaerobic respiration. [15]

Similarities

- ref. to glycolysis
- glucose broken down to pyruvate and 2 ATP and 2 NADH
- substrate-level phosphorylation
- reduction of NAD to NADH
- occurs in the cytoplasm / cytosol
- ref. to enzyme-catalyzed reaction

Differences

- ref. to involves glycolysis, link reaction, Krebs cycle and oxidative phosphorylation
- ref. to presence / absence of oxygen
- ref. to 36 / 38 ATP vs 2 ATP
- ref. to products of respiration in glycolysis
- ref. to products of aerobic vs anaerobic respiration
- ref. to final electron acceptor
- ref. to location
- ref. to substrate level phosphorylation vs oxidative phosphorylation
- ref. to different roles of NADH
- ref. to chemiosmosis occurring or lack of it
- ref. to oxidative decarboxylation or lack of it
- ref. to complete / incomplete oxidation of glucose

(b) Plants are important indicators used by scientists to study the effects of environmental stress caused by climate change.

Discuss the effects of climate change on the rich biodiversity of plants in the tropics. [10]

- ref. to definition of climate change
- ref. to definition of biodiversity
- ref. to increase in frequency of extreme weather events
- ref. to shifts in geographic ranges
- ref. to reduction in amount and availability of habitat / biomes
- ref. to loss of ecosystem diversity
- ref. to plants selection against plants unable to adapt
- ref. to increased temperature leading to higher environmental stress on plants
- ref. to shifts in flowering time of plants
- ref. to impact on insects
- ref. to climate change decreasing biodiversity in the tropics
- ref. to resulting in extinction of plant species
- ref to decrease gene pool

